

## Documentation:

- We went and found the private key in the folder that was shared with us
- We then copied the path and ssh to be able to connect to the student01 virtual machine
- We did this by using the personal computers shell not the powershell in azure
- We then used nmap -sn 10.10.10.4/24 and found 1 host which was us
- We then used nmap -sv -sc -p- 10.10.20.4 to scan the ports

```
(student01@kali)~$ nmap -sv -sc -p- 10.10.20.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 19:02 +0000
Stats: 0:01:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 58.84% done; ETC: 19:04 (0:00:49 remaining)
Stats: 0:02:24 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 52.94% done; ETC: 19:05 (0:00:35 remaining)
Nmap scan report for victim.internal.cloudapp.net (10.10.20.4)
Host is up (0.0016s latency).
Not shown: 65518 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          FileZilla ftpd 1.12.6
|_ftp-anon: got code 503 "Use AUTH first.".
|_ftp-syst:
|_SYST: UNIX emulated by FileZilla.
|_ssl-cert: Subject: commonName=filezilla-server self signed certificate
|_Not valid before: 2026-07-29T20:30:04
|_Not valid after: 2027-07-30T20:35:04
|_tls-alpn:
|_ftp
80/tcp    open  http         Microsoft IIS httpd 10.0
|_http-methods:
|_Potentially risky methods: TRACE
|_http-server-header: Microsoft-IIS/10.0
|_http-title: IIS Windows Server
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
1433/tcp  open  ms-sql-s     Microsoft SQL Server 2019 15.00.2000.00; RTM
|_ssl-date: 2026-08-04T19:05:44+00:00; +1s from scanner time.
|_ms-sql-info:
|_10.10.20.4\MSSQLSERVER:
|_Instance name: MSSQLSERVER
|_Version:
|_name: Microsoft SQL Server 2019 RTM
|_number: 15.00.2000.00
|_Product: Microsoft SQL Server 2019
|_Service pack level: RTM
|_Post-SP patches applied: false
|_TCP port: 1433
|_Clustered: false
|_ssl-cert: Subject: commonName=SSL_Self_Signed_Fallback
|_Not valid before: 2026-07-29T20:32:51
```

```
Not valid before: 2026-07-29T20:32:51
Not valid after: 2026-07-29T20:32:51
ms-sql-ntlm-info:
10.10.20.4\MSSQLSERVER:
Target_Name: victim
NetBIOS_Domain_Name: victim
NetBIOS_Computer_Name: victim
DNS_Domain_Name: victim
DNS_Computer_Name: victim
Product_Version: 10.0.20348
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
|_rdp-ntlm-info:
|_Target_Name: victim
|_NetBIOS_Domain_Name: victim
|_NetBIOS_Computer_Name: victim
|_DNS_Domain_Name: victim
|_DNS_Computer_Name: victim
|_Product_Version: 10.0.20348
|_System_Time: 2026-08-04T19:05:05+00:00
|_ssl-cert: Subject: commonName=victim
|_Not valid before: 2026-07-28T18:49:40
|_Not valid after: 2027-01-27T18:49:40
|_ssl-date: 2026-08-04T19:05:44+00:00; +1s from scanner time.
5985/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
47001/tcp open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
49664/tcp open  msrpc        Microsoft Windows RPC
49665/tcp open  msrpc        Microsoft Windows RPC
49666/tcp open  msrpc        Microsoft Windows RPC
49667/tcp open  msrpc        Microsoft Windows RPC
49668/tcp open  msrpc        Microsoft Windows RPC
49669/tcp open  msrpc        Microsoft Windows RPC
49671/tcp open  msrpc        Microsoft Windows RPC
50766/tcp open  msrpc        Microsoft Windows RPC
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
smb2-time:
date: 2026-08-04T19:05:09
start_date: N/A
```

- We then used were attempting to find the versions for all ports
- We used the command searchsploit "FileZilla Server 1.12.6"
- searchsploit "Microsoft IIS 10.0"
- searchsploit "Microsoft SQL Server 2019"
- searchsploit "Windows Server 2022"

- searchsploit "Microsoft HTTPAPI 2.0"

```
(student01@kali)~$ searchsploit "FileZilla Server 1.12.6"
searchsploit "Microsoft IIS 10.0"
searchsploit "Microsoft SQL Server 2019"
searchsploit "Windows Server 2022"
searchsploit "Microsoft HTTPAPI 2.0"
Exploits: No Results
Shellcodes: No Results
Exploits: No Results
Shellcodes: No Results
```

| Exploit Title                                                      | Path             |
|--------------------------------------------------------------------|------------------|
| Microsoft Windows SQL Server - Remote Denial of Service (MS03-031) | windows/dos/65.c |

```
Shellcodes: No Results
```

| Exploit Title                                                                     | Path                     |
|-----------------------------------------------------------------------------------|--------------------------|
| CitectSCADA ODBC Server - Remote Stack Buffer Overflow (Metasploit)               | windows/remote/6387.rb   |
| CoreFTP Server build 725 - Directory Traversal (Authenticated)                    | windows/remote/50852.txt |
| EaseUS Data Recovery - 'enserver.exe' Unquoted Service Path                       | windows/local/50886.txt  |
| Easy Chat Server 3.1 - Remote Stack Buffer Overflow (SEH)                         | windows/remote/50999.py  |
| Emerson PAC Machine Edition 9.80 Build 8695 - 'TrapiServer' Unquoted Service Path | windows/local/50745.txt  |
| Microsoft Windows Server 2000 - 'telnet.exe' NTLM Authentication                  | windows/remote/20222.cpp |
| Microsoft Windows Server 2000 - RunAs Service Denial of Service                   | windows/dos/21099.c      |
| Microsoft Windows Server 2000 - RunAs Service Named Pipe Hijacking                | windows/local/21069.c    |
| Oracle Weblogic Server 14.1.1.0.0 - Local File Inclusion                          | windows/remote/50688.txt |
| Sambar Server 4.3/4.4 Beta 3 - Search CGI                                         | windows/remote/20223.txt |
| TCQ - IfeCProteccioAppServer.exe - Unquoted Service Path                          | windows/local/50902.txt  |
| TermTalk Server 3.24.0.2 - Arbitrary File Read (Unauthenticated)                  | windows/remote/50638.txt |
| TRIGONE Remote System Monitor 3.61 - Unquoted Service Path                        | windows/local/50633.txt  |
| TYPSoft FTP Server 0.7.x - FTP Server Remote Denial of Service                    | windows/dos/20228.pl     |
| WiFi Mouse 1.7.8.5 - Remote Code Execution(v2)                                    | windows/remote/50972.py  |
| Wing FTP Server 4.3.8 - Remote Code Execution (RCE) (Authenticated)               | windows/remote/50720.py  |

```
Shellcodes: No Results
Exploits: No Results
Shellcodes: No Results
```

- This didn't show the version so we ran a different command nmap -sV 10.10.20.4

```
(student01@kali)~$ nmap -sV 10.10.20.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 19:23 +0000
Nmap scan report for victim.internal.cloudapp.net (10.10.20.4)
Host is up (0.0014s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          FileZilla ftpd 1.12.6
80/tcp    open  http         Microsoft IIS httpd 10.0
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
1433/tcp  open  ms-sql-s     Microsoft SQL Server 2019 15.00.2000
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
5985/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.70 seconds
```

- The vulnerability seemed to be microsoft sql server 2019
- We then ran the Nmap --script vuln 10.10.20.4

```
(student01@kali)~$ nmap --script vuln 10.10.20.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 19:32 +0000
Stats: 0:02:05 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.46% done; ETC: 19:34 (0:00:01 remaining)
Nmap scan report for victim.internal.cloudapp.net (10.10.20.4)
Host is up (0.0011s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-enum:
| /admin/: Possible admin folder
| /admin/index.html: Possible admin folder
| /Admin/: Possible admin folder
|_ /backup/: Possible backup
|_http-dombased-xss: Couldn't find any DOM based XSS.
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
1433/tcp   open  ms-sql-s
|_tls-ticketbleed: ERROR: Script execution failed (use -d to debug)
3389/tcp   open  ms-wbt-server
5985/tcp   open  wsmn
Host script results:
|_smb-vuln-ms10-054: false
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
Nmap done: 1 IP address (1 host up) scanned in 143.51 seconds
```

- We then ran curl <http://10.10.20.4/admin/> and found that it was open and should not be public
- The ports that we found vulnerable were port 80, port 1443, port 139

#### Part 4:

For port 1443 updating sql server since its using an outdated version. For port 80 closing the port if it isn't needed. If it is needed keeping the web server updated and monitoring would be the best step. For port 139 if it isn't needed, close it if it is needed, restrict access and find what is listening on the port.

#### Reflection Questions:

1. The easiest piece of information to find that would help an attacker would be finding out that port 1443 had an outdated version of sql running. Also the information for all the versions being easily accessible. This was exposed because open ports can reveal the services running on a computer including the software type, version, and system information.
2. The first change I would make is updating the sql server because the system is outdated and if it hasn't been patched it can be used for a data breach. Compromising the CIA triad through multiple ways.
3. Port 443 the HTTP port being open is a medium risk if it isn't being used then it is an unnecessary attack surface. It is basically giving attackers a way to have access on all services listed on that port and some of the services may have vulnerabilities. It can also reveal information about the software and version of the web server or application. It is the same with Port 1443 since the sql server is outdated it is leaving an open door for any attacker to walk through to gain information of any services or information kept within it. Port 139 is a low risk vulnerability because it was an inconclusive result when checking for vulnerabilities but should still be recorded and monitored because if it isn't needed, since it is typically only used for legacy NetBIOS support, it can reveal information such as the computer name, domain/workgroup, operating system information etc.